Venkata Lakshmi Parimala Pasupuleti

Sr. Software Engineer · Founding Engineer · Multifamily PropTech · Java/Spring Boot · React · Python · Kubernetes · Docker · Production Systems Ownership · 9yr

+1-703-261-5410 | venkatalakshmiparimalap@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti | github.com/venkatalakshmiparimala | Arlington, Virginia, USA

PROFESSIONAL SUMMARY

Senior software engineer with 9+ years of production experience and 7 years as the sole long-term owner of business-critical systems in multifamily real estate - the exact founding engineer profile RangeWater is building for. At Entrata I owned the HUD income-limit compliance pipeline end to end for nearly 7 years with no platform team behind me: designed it, shipped it, monitored it, resolved production incidents, and maintained zero compliance violations across the full ownership period. This wasn't a slice of a feature team - it was a system the business depended on monthly, built in Java and PHP with PostgreSQL, RabbitMQ async processing, Docker microservices, and Kubernetes orchestration. I know the property management domain from the inside: leasing workflows, eligibility calculations, rent and HAP determinations, resident-facing data - the same operational context RangeWater's platform serves. On the AI side: LeaseI (8-agent LangGraph compliance platform) and AgentProbe (production evaluation toolkit) show I already use AI as a force multiplier, not a novelty. I ship complete solutions, own them through production, and treat a blank page as a starting line. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

Languages & Frameworks: Java, Spring Boot, Python (9yr), PHP/Laravel, JavaScript, TypeScript, React.js, Angular, FastAPI, REST APIs, gRPC-adjacent service design, SQL, PL/SQL, Bash

Infrastructure & Production Systems: Kubernetes (K8s), Docker, CI/CD (GitHub Actions), AWS (S3, EC2, RDS, Glue, Lambda), Google Cloud Storage (GCP), Azure, RabbitMQ, Microservices, Distributed Systems, Event-Driven Architecture, Async Processing, Idempotency, API Design, Data Modeling, Schema Design & Migrations, Performance Engineering, Query Optimization, Scalable Systems, On-Call Production Ownership

Data, Observability & Security: PostgreSQL, MySQL, Microsoft SQL Server, Oracle, NoSQL, Libsodium encryption, PHI/HIPAA-compliant data pipelines, Structured Logging, Monitoring, Alerting, Incident Response, Root Cause Analysis, Data Integrity, Secure Coding, ETL/Data Pipelines, Third-Party API Integration

AI & Developer Tooling: LangGraph, LangChain, RAG Pipelines, LLM Evaluation, Agentic Systems, HITL Guardrails, ChromaDB, Anthropic API, OpenAI API, AI Coding Tools (Claude Code), Agile/Scrum, Technical Mentoring, Code Reviews

EXPERIENCE

Magizhs Tech Solution
May 2025 - Aug 2025

AI/ML Engineer - 0→1 Sole Technical Owner

- Took a production AI system from zero to deployed as sole technical owner - designed architecture, built backend services, implemented data pipeline, evaluation infrastructure, and CI validation gate; delivered in weeks. Direct analog to the founding engineer ownership model RangeWater is hiring for.

George Mason University
Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Built multi-source ML pipelines (Python, R^2=0.977) on AWS and GCP; applied anomaly detection across 3 research pipelines; mentored 600+ students in Python, SQL, and distributed systems.

Entrata - Multifamily Property Management SaaS
Sep 2018 - Aug 2024

Senior Software Engineer - Sole Long-Term Owner, Production Systems · Lehi, UT (Remote)

- **Founding engineer ownership for 7 years** - owned the HUD income-limit compliance pipeline end to end with no platform team: annual federal data ingestion and normalization (few thousand source rows → 180,000 domain records powering LIHTC eligibility, rent, HAP, and tax-credit calculations for 100K+ housing units); designed it, shipped it, monitored it, and maintained zero compliance violations across full tenure. Deployed it, resolved every production incident, and made sure failures didn't recur.
- **Distributed backend & infrastructure** - RabbitMQ event-driven async processing (retry logic, idempotency guarantees) for gross-rent updates across thousands of tenant records; Docker microservices redesign (50% scalability improvement); Kubernetes orchestration; 200+ PostgreSQL/MySQL query optimizations (30% performance gain); GitHub Actions CI/CD with comprehensive automated test suites.
- **Full-stack delivery, security & integrations** - shipped 20+ production modules (Java, PHP, Python, JavaScript, React.js) across 120+ Agile sprints; led AWS S3 migration with Libsodium-encrypted ETL pipelines (100% PHI compliance, zero security incidents); third-party federal API integrations (HUD data, document systems); built internal debugging tools reducing triage time 40%; mentored 5+ engineers; 300+ code reviews.

Entrata India - Pune
Jan 2017 - Sep 2018

Associate Software Engineer → Trainee

- Built property management modules (Java, PHP, PostgreSQL); data pipeline for 1.6M HUD/Tax Credit records; AWS S3 migration resolving 90% of legacy storage constraints.

PROJECTS

Leasel: Production Agentic Platform - PropTech Compliance, Sole Owner

Agentic · LangGraph · RAG · HITL · PropTech · Production · Full-Stack

Python · LangGraph (8-agent) · FastAPI · TF-IDF RAG · SQLite · React · Anthropic API | github.com/venkatalakshmiparimala/Lease_complaince_rag

End-to-end production AI platform built and owned alone - the founding engineer model in action. Backend services, data pipeline, RAG retrieval layer, agentic orchestration, React frontend, and evaluation infrastructure all designed and shipped by one engineer. 8-agent LangGraph pipeline (Triage → Retrieval → Configuration → Incident → Reasoning → Validation → Escalation) automating property compliance investigations with HITL guardrails. 93.6% precision, 87.9% F1 across 1,240 test cases. Core rule enforced in code: LLM reasons, deterministic logic makes every final regulated call. Demonstrates AI as a force multiplier for production engineering, not a demo.

AgentProbe: AI Evaluation & Developer Productivity Platform

Production Tooling · CI/CD · FastAPI · React · TypeScript · Open Source

Python · FastAPI · React.js · TypeScript · LangGraph · ChromaDB · GitHub Actions · REST APIs | github.com/venkatalakshmiparimala/AgentProbe

Production developer tooling platform built alone: 15-endpoint FastAPI backend, React.js + TypeScript frontend, CI-integrated regression gate (GitHub Actions) blocking quality regressions before deployment, distributed tracing across all system operations, and structured observability throughout. Demonstrates the end-to-end ownership mindset - backend, infrastructure, frontend, and monitoring all shipped without a team. Used to evaluate Leasel across 1,240+ test cases with zero regressions.

AI on FHIR: Full-Stack Cloud Application - Regulated Domain

Full Stack · AWS Lambda · GCP · HIPAA · REST APIs · Docker · PostgreSQL

Python · Flask · React.js · TypeScript · Docker · AWS Lambda · GCP · PostgreSQL · REST APIs | github.com/venkatalakshmiparimala/AI_on_FHIR

Full-stack AI application on cloud infrastructure for a regulated domain - LLM-powered NLP translating natural-language queries (EN/ES) into FHIR API calls; HIPAA-compliant RBAC with full audit trail; AWS Lambda + GCP backend; Docker containerization; PostgreSQL data layer with complete observability. Demonstrates shipping polished, complex full-stack systems in regulated environments where data integrity and security are non-negotiable.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017